

AI-POWERED HEALTHCARE ASSISTANT

**Predicting Diseases *Generating Reports *Answering patient Queries*

These technologies Work together to deliver intelligent , accurate and personalized healthcare

ARTIFICIAL INTELLIGENCE (AI)

The broad field of building intelligent systems that can perform tasks normally requiring human intelligence.

ROLE IN HEALTHCARE ASSISTANT

- *Improves efficiency
- *Provides overall framework
- *Enable automation healthcare tasks.

MACHINE LEARNING (ML)

A Subset of AI that uses algorithms to learn patterns from data and make predictions or decisions without explicit programming

ROLE IN HEALTHCARE ASSISTANT

- *Analysis of structured data
- *Predicts risk of disease
- *Supports clinical decisions

DEEP LEARNING(DL)

A Subset of ML that uses multi layer neural networks to learn complex patterns from large amount of data

ROLE IN HEALTHCARE ASSISTANT

- *Analysis of unstructured data
- *Detects abnormalities
- *Improves accuracy

GENERATIVE AI

A Subset of DL that can generate new content such as image and medical data

ROLES IN HEALTHCARE ASSISTANT

- *Generates medical reports
- *Summarize long documents
- *Creates synhetic data

LARGE LANGUAGE MODELS (LLMs)

A type of generative ai trained on vast text data understand human like structure.

ROLES IN HEALTHCARE ASSISTANT

- *Explains symptoms
- *Powers the patient chatbot
- *Assists doctors by research

INTEGRATED RESULTS

Combining AI,ML,DL,GENERATIVE AI,ANDLLMs enables the healthcare assistant to predict disease ,genrate reports and answer queries with high accuracy personalization and scalability.

AI-BASED LEARNING PLATFORM

ARTIFICIAL INTELLIGENCE (AI)

Overall system control and decision making integrates all.

MACHINE LEARNING (ML)

Analyses student behaviour, provides personalised recommendation

DEEP LEARNING (DL)

Understands complex data improves accuracy in prediction and analysis

GENERATIVE AI (GEN AI)

Generates new content such as notes, summaries and presentation

LLMs

Enables natural language interaction acts as a intelligent tutor and doubts

KEY FEATURES OF THE PLATFORM

PERSONALIZED RECOMMENDATION

AUTOMATED GRADING

CONTENT GENERATION

INTELLIGENT TUTOR

LEARNING OUTCOMES

BETTER LEARNING

HIGHER ENGAGEMENT

IMPROVED PERFORMANCE

WHAT IF ALL FIVE TECHNOLOGIES ARE NECESSARY ?

- * AI PROVIDES OVERALL INTELLIGENCE AND INTEGRATES EVERY COMPONENT
- * ML LEARNS FROM DATA TO GIVE ADAPTIVE EXPERIENCE
- * DL HANDLES COMPLEX PATTERNS ALONE CANNOT PROCESS EXPERIENCE